Pre-Algebra

Նախահանրահաշիվ

COURSE CODE	MTH-PA
PROGRAM	Middle School Mathematics · Միջին դպրոցի մաթեմատիկա
LEVEL	Bridge course · typically Grades 7–9
DURATION	60–72 instructional hours (approx. 54 lessons)
PREREQUISITE	Grade 6 Mathematics or equivalent, or placement by assessment. Suitable for any student entering Algebra I with gaps.
LEADS TO	Algebra I (MTH-A1)

COURSE DESCRIPTION

Դասընթացի նկարագրություն

Pre-Algebra is a bridge course, not a grade level. It exists for the student who has finished middle school arithmetic but is not yet ready for the pace and abstraction of Algebra I — and for the student who is entering Algebra I from a different school system and needs the gaps found by the placement assessment closed efficiently. The course consolidates rational-number fluency, proportional reasoning, and the language of algebra, then takes the student through linear equations and functions to the threshold of Algebra I. It is deliberately narrower and deeper than a grade-level course.

TEACHING APPROACH

Դասավանդման մոտեցումը Վարժարանում

Most students who struggle in Algebra I do not struggle with algebra. They struggle with negative numbers, with fractions, with the meaning of the equals sign, and with reading a symbol as a quantity. Վարժարան designed this course around that observation: it spends its time on the small number of ideas that actually cause Algebra I failure, and it does so with a diagnostic-first structure. The placement assessment determines which units a given student needs, and a student with a clean profile in a given area may skip that unit entirely. Nothing here is taught for completeness; everything is taught because its absence causes a specific downstream failure.

PLACEMENT AND READINESS INDICATORS

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The placement assessment checks each indicator below. A student missing two or more is enrolled with a remediation plan attached to the personalized curriculum.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Performs the four operations with fractions and decimals	If absent, begin with Unit 1 in full; otherwise Unit 1 may be compressed or skipped.
Orders negative numbers and understands absolute value	Determines whether Unit 2 is needed at full length.
Solves a one-step equation and checks by substitution	Determines the entry point into Unit 4.
Interprets a ratio or a rate in context	Determines whether Unit 3 is needed; the most common gap in students arriving from other systems.

COURSE LEARNING OUTCOMES

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On completion of this course the student will be able to:

1. Compute fluently and confidently with all rational numbers in fraction, decimal, and percent form.
2. Apply the order of operations and the properties of operations correctly and flexibly.
3. Work with integer exponents, square roots, and scientific notation.
4. Reason proportionally and solve percent, rate, and scale problems.
5. Read, write, and manipulate algebraic expressions using correct vocabulary.
6. Solve multi-step linear equations and inequalities, including variables on both sides.
7. Translate between verbal situations and algebraic models in both directions.
8. Understand the function concept and use function notation.
9. Graph linear relationships and interpret slope and intercept in context.
10. Apply the Pythagorean theorem and compute area, surface area, and volume.
11. Approach an unfamiliar problem with a deliberate strategy rather than by guessing at an operation.

COURSE UNITS

Դասընթացի բաժիններ

Each unit below lists the lesson-level topics a teacher delivers, the objectives that define success, the observable mastery criteria used to update the student's progress profile, and the misconceptions this unit reliably produces together with the recommended teacher response.

UNIT 1 · 8-9 LESSONS

Rational Number Fluency

Ռացիոնալ թվերի սահուն կիրառում

TOPICS COVERED

- The four operations with fractions, revisited for fluency and speed
- The four operations with decimals
- Converting among fractions, decimals, and percents
- Order of operations with nested grouping and exponents
- Commutative, associative, distributive, identity, and inverse properties by name
- Using properties to compute efficiently rather than mechanically
- Estimation and reasonableness checking
- GCF, LCM, and prime factorization
- Choosing the most convenient form of a number for a given task

LEARNING OBJECTIVES

1. Compute with any rational number in any form, choosing the most convenient representation.
2. Name the property justifying each step of a computation.
3. Estimate before computing and use the estimate to detect errors.
4. Convert fluently among fraction, decimal, and percent forms.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Computes with rational numbers at 92% accuracy across all forms.
- Names the property used in 8 of 10 prompted steps.
- Detects a planted error by estimation in 4 of 5 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Order of operations is applied as a rigid left-to-right sequence of the letters in a mnemonic.	Multiplication and division share a level and are performed left to right, as do addition and subtraction. Use contrasting pairs such as $8 \div 4 \times 2$ until the level structure is clear.
Percent is treated as a separate number type rather than a form of a rational number.	Convert in both directions in the same lesson and require the student to state which form is easiest for the task at hand.

TEACHING NOTE

This unit is skipped entirely for students whose placement assessment shows clean rational-number fluency. It exists to be diagnostic, not obligatory.

UNIT 2 · 8–9 LESSONS

Integers, Exponents, and Roots

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TOPICS COVERED

- The number line extended and the meaning of a negative quantity
- Adding and subtracting integers with models
- Subtraction as adding the opposite
- Multiplying and dividing integers and justifying the sign rules
- Operations with negative fractions and decimals
- Absolute value as distance
- Integer exponents and the exponent properties
- Zero and negative exponents
- Square roots and cube roots; perfect squares to 225
- Estimating a square root between consecutive integers
- Rational and irrational numbers
- Scientific notation and computing with it

LEARNING OBJECTIVES

1. Compute with negative numbers in all forms fluently.
2. Justify the sign rules from patterns or from the distributive property.
3. Simplify expressions using integer exponent properties.
4. Estimate a square root and place an irrational number on a number line.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Computes with negatives at 92% accuracy on interleaved problem sets.
- Simplifies exponent expressions at 90% accuracy.
- Estimates square roots to the nearest tenth in 4 of 5 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Sign errors appear in long computations even when the rules are known in isolation.	This is a working-memory problem, not a knowledge problem. Require intermediate steps to be written; forbid mental multi-step sign handling until accuracy is stable.
−x is assumed to be a negative number.	It is the opposite of x, which is positive when x is negative. Substitute x = −3 explicitly. This misconception causes real damage in Algebra I.

TEACHING NOTE

Interleaved practice is essential here. A student who scores well on a page of multiplication-only sign problems has not demonstrated anything; mix the operations from the first assignment.

UNIT 3 · 8-9 LESSONS

Ratio, Proportion, and Percent

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TOPICS COVERED

- Ratios, rates, and unit rates
- Ratio tables, tape diagrams, and double number lines
- Proportional relationships in tables, graphs, equations, and descriptions
- The constant of proportionality and its interpretation
- Solving proportions, with cross-multiplication introduced last and justified
- Non-proportional situations and how to recognize them
- Percent of a number; finding the whole; finding the percent
- Percent increase and decrease; markup, discount, tax, tip
- Simple interest
- Scale drawings, maps, and models
- Unit conversion by rate reasoning

LEARNING OBJECTIVES

1. Decide whether a relationship is proportional and justify from any representation.
2. Solve a proportion by scaling, by unit rate, and by cross-multiplication, and explain why all three work.
3. Solve multi-step percent problems including successive changes.
4. Convert units within and between systems by multiplying by rates equal to one.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Classifies relationships correctly in 9 of 10 trials with justification.
- Solves multi-step percent problems at 88% accuracy.
- Performs multi-step unit conversions correctly in 8 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Cross-multiplication is applied to any pair of fractions, including in addition problems.	It is a technique for solving a proportion, not a general operation. Introduce it only after scaling and unit-rate methods are secure.
A proportion is set up with mismatched units across the two ratios.	Label every quantity with its unit inside the proportion. Mismatched labels reveal the setup error before any arithmetic occurs.

TEACHING NOTE

Proportional reasoning is the most transferable idea in this course and the gap most often found in students arriving from other school systems. Even a strong placement profile warrants a diagnostic pass through this unit.

UNIT 4 · 9-10 LESSONS

The Language of Algebra

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TOPICS COVERED

- Variables as quantities that vary, and as unknowns to be found
- Vocabulary: term, coefficient, constant, factor, like terms, expression, equation
- Translating verbal phrases into expressions and expressions into words
- Evaluating expressions at given values, including negatives
- Combining like terms
- The distributive property with variables, including negative multipliers
- Expanding and factoring simple linear expressions
- Determining whether two expressions are equivalent
- The meaning of a solution to an equation
- Solving one-step and two-step equations
- Solving equations with variables on both sides
- Equations requiring expansion first
- Checking solutions by substitution
- Solving and graphing linear inequalities
- Reversing the inequality when multiplying by a negative
- Writing equations and inequalities from word problems

LEARNING OBJECTIVES

1. Use algebraic vocabulary precisely in speech and in writing.
2. Translate fluently in both directions between verbal descriptions and algebraic symbols.
3. Solve multi-step linear equations and inequalities and verify solutions.
4. Write an equation or inequality modelling a described situation and interpret the solution.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Solves multi-step equations at 90% accuracy with checking shown.
- Translates in both directions correctly in 9 of 10 trials.
- Uses vocabulary correctly without prompting in 8 of 10 observed instances.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
A variable is treated as a label for an object rather than as a number.	The single most damaging misconception in early algebra. Insist that the variable represents *the number of* things, and substitute values often to keep it numeric.

MISCONCEPTION	RECOMMENDED RESPONSE
Like terms are combined incorrectly: 3x + 2 becomes 5x.	Terms with different variable parts cannot be combined. Use a concrete analogy — three x's and two units are not five of anything — and substitute a value to demonstrate.
Distributing a negative multiplier changes only the first sign.	Write the multiplier with its sign attached and draw arrows to every term. Then check by substituting a value into both forms.

TEACHING NOTE

This is the core unit of the course and should receive the most time. A student who leaves it with precise vocabulary and reliable equation-solving will pass Algebra I; one who does not, will not, regardless of what else they know.

UNIT 5 · 8-9 LESSONS

Linear Relationships and Functions

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TOPICS COVERED

- The coordinate plane and graphing ordered pairs in four quadrants
- Tables, graphs, and equations as three views of one relationship
- The definition of a function and testing for it
- Function notation and reading f(x)
- Rate of change and slope, with units
- Slope from two points, a graph, a table, and an equation
- Intercepts and their meaning in context
- Slope-intercept form and graphing from it
- Writing the equation of a line from a graph, a table, or a description
- Linear versus nonlinear relationships
- Systems of two linear equations by graphing and substitution
- Interpreting a solution in context

LEARNING OBJECTIVES

1. Move fluently among table, graph, and equation for a linear relationship.
2. Apply the definition of a function and use function notation correctly.
3. Compute slope from any representation and interpret it with units in context.
4. Write the equation of a line from any adequate description and solve a simple system.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Moves among representations correctly in 9 of 10 tasks.
- Interprets slope and intercept in context with units in 9 of 10 trials.
- Solves systems by graphing and substitution at 88% accuracy.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Slope is memorized as rise over run with no meaning attached.	Require a contextual sentence with units for every slope computed. A number with no interpretation is marked incomplete.
f(x) is read as multiplication.	Correct immediately and permanently. The cost of allowing this to persist is very high in Algebra II.

TEACHING NOTE
Ending the course with functions rather than with geometry is deliberate: it hands the student directly to Algebra I Unit 1 with the right vocabulary already in place.

UNIT 6 · 6–7 LESSONS

Geometry and Measurement Essentials

Երկրաչափության և չափման հիմունքներ

TOPICS COVERED

- Angle relationships: complementary, supplementary, vertical, adjacent
- Angles formed by parallel lines and a transversal
- Triangle angle sum and the exterior angle theorem
- The Pythagorean theorem, its converse, and applications
- Distance between two points on the coordinate plane
- Area and perimeter of triangles, quadrilaterals, and circles
- Area of composite figures
- Surface area and volume of prisms, cylinders, cones, and spheres
- Similar figures and scale factor; how area and volume scale
- Unit analysis and dimensional reasoning

LEARNING OBJECTIVES

1. Solve for unknown angles by writing and solving an equation with a stated reason.
2. Apply the Pythagorean theorem in two and three dimensions and derive the distance formula.
3. Compute area, surface area, and volume for standard figures with correct units.
4. Explain how area and volume scale under a dilation.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Solves unknown-angle problems with reasons in 9 of 10 trials.
- Applies the Pythagorean theorem at 90% accuracy.
- Computes area, surface area, and volume with correct units in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Formulas are memorized without dimensional understanding, so area formulas are used for volume.	Reason from units. A volume must produce cubic units; a formula that cannot is the wrong formula.
Under a scale factor of 3, volume is assumed to triple.	Volume scales by the cube of the factor. Build a 1-cube and a 3-cube from unit cubes and count.

TEACHING NOTE

Unit analysis introduced here is one of the most powerful error-catching tools a student can own, and it costs almost nothing to teach. Require units on every answer for the remainder of the course.

ASSESSMENT AND MASTERY POLICY

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ASSESSMENT	WHEN	WHAT IT MEASURES
Diagnostic Placement Assessment	Before enrollment	Comprehensive: determines which units the student needs and in what order. Units with a clean profile may be skipped entirely.
Unit Check	End of each unit · 40 min	Unit mastery criteria; a failed check returns the student to targeted lessons rather than advancing.
Fluency Probe	Every 2nd lesson · 5 min	Rational-number and integer computation, interleaved.
Algebra Readiness Assessment	After Unit 4	Equation-solving, translation, and vocabulary; the gate for Algebra I entry.
End-of-Course Assessment	After Unit 6	All course outcomes; produces a formal Algebra I readiness recommendation for the parent.

Վարժարան reports mastery, not grades. Each topic in the student's profile is marked **Mastered**, **Developing**, or **Needs Review** against the criteria stated in each unit above. A topic is marked Mastered only when the criterion is met on two separate occasions at least one week apart — this prevents short-term recall from being recorded as durable learning. Topics marked Needs Review are automatically re-entered into homework and into the opening minutes of subsequent lessons until they are re-mastered.

PACING OPTIONS

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TRACK	CADENCE	EXPECTED COMPLETION
Standard	1 lesson/week, 60 min	Full course in approximately 40–45 weeks.
Accelerated	2 lessons/week	Full course in approximately 22–26 weeks.
Targeted (diagnostic-driven)	2 lessons/week	Only the units indicated by the placement assessment; typically 10–20 weeks.
Summer Algebra Bridge	3 lessons/week, 8 weeks	Units 2, 3, and 4 — designed for students entering Algebra I in the autumn.

Pacing is assigned after the placement assessment and reviewed at the mid-year assessment. A student who is meeting mastery criteria early may be moved to the accelerated track; a student who is not may be moved to intensive without any change in the course content itself.

HOMEWORK AND INDEPENDENT PRACTICE

Տնային աշխատանք և ինքնուրույն պարապմունք

Thirty-five to forty-five minutes per lesson. Assignments are interleaved by design: each contains current content, one problem from the previous unit, and one from an earlier one. Every equation solved must be checked by substitution, and every applied answer must carry units and a sentence of interpretation. Students maintain an error log; because this course exists to close specific gaps, knowing precisely which errors recur is more valuable here than in any other course in the catalog.

PROGRESS MILESTONES REPORTED TO PARENTS

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These are the achievements the parent portal announces as the student reaches them. They replace attendance counts as the primary measure of progress.

- Computes fluently with all rational numbers in every form.
- Justifies the integer sign rules rather than reciting them.
- Applies exponent properties and estimates square roots.
- Recognizes proportional relationships and solves multi-step percent problems.
- Uses algebraic vocabulary precisely.
- Translates fluently between words and algebraic symbols.
- Solves multi-step equations and inequalities with variables on both sides.
- Understands the function concept and uses function notation.
- Computes and interprets slope with units in context.
- Applies the Pythagorean theorem and computes area, surface area, and volume.
- Completed Pre-Algebra — formally assessed as ready for Algebra I.

MATERIALS AND RESOURCES

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- Scientific calculator (graphing calculator not required for this course).
- Two-color counters and an extended number line for integer work.
- Graphing tool for the linear relationships unit.
- Compass, protractor, and ruler.
- Վարժարան Pre-Algebra problem bank — approximately 1,100 items organized by diagnostic gap rather than by lesson.

Վարժարան · Varzharan — Exceptional Armenian educators. American academic standards. Personalized education. | This syllabus defines the standard course. Every enrolled student receives a personalized version of it, sequenced from their placement assessment and revised as their progress profile changes.